# Seamus R. Howrigan

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## PROFESSIONAL SUMMARY

Mechanical Engineering graduate of the University of Vermont with hands-on R&D and test engineering experience across aerospace and industrial applications. Skilled in NASTRAN-based FEA structural analysis, data acquisition system development, dynamic and static seal testing, and laboratory instrumentation. Currently contributing as an R&D Machinery Lab Technician at Flexaseal Engineered Seals and Systems.

## EDUCATION

**University of Vermont** Burlington, VT
*Bachelor of Science in Mechanical Engineering.* *May 2026*
Relevant Coursework: Compressible Flow, Control Systems, Mechatronics

## EXPERIENCE

**R&D Machinery Lab Technician** June 2026 – Present
*Flexaseal Engineered Seals and Systems* *Williston, VT*
- Execute dynamic testing for product qualification, performance analysis, and testing-to-limits on mechanical seal assemblies.
- Schedule and coordinate tests, assemble test cartridges, and perform data reduction and cataloging to support R&D decisions.
- Maintain laboratory equipment and instrumentation assets, including machinery and sensor troubleshooting.
- Assist with test rig design and build, and support miscellaneous engineering, measurement, and fabrication tasks.

**R&D Intern** June 2024 – August 2025
*Flexaseal Engineered Seals and Systems* *Williston, VT*
- Reverse-engineered competitor mechanical seals and drawings to support customer repair workflows and inform R&D teams.
- Built and configured NASTRAN FEA models of manufactured parts for structural analysis in ultra-high pressure applications.
- Refined a DAQFactory data-logging dashboard to display reduced data from digital instrumentation and VFDs per test seal parameters.
- Conducted high-pressure static integrity tests of job seals with nitrogen and water, recording data to drive R&D decisions.

**DBF Testing Team Lead / President** May 2023 – May 2026
*American Institute of Aeronautics and Astronautics, UVM Chapter* *Burlington, VT*
- Led a flight and component test program encompassing approximately half a dozen flight tests and multiple component tests, directly informing competition aircraft design and validation.
- Enforced UVM, State of Vermont, and FAA regulations for all flight and powered test operations.
- Engaged industry and academic partners to grow club membership and taught engineering principles through aerospace competitions.

**Intern Team Member** February 2021 – August 2021
*Beta Technologies* *South Burlington, VT*
- Supported the Subscale Model Team by designing foam molds for composite layups and repairing scaled drone prototypes.
- Sorted and organized telemetry data from test flights to support analysis efforts.

**Peer Tutor, Introduction to Python** August 2023 – December 2023
*University of Vermont* *Burlington, VT*
- Guided students through Python programming fundamentals, developing problem-solving skills through targeted instruction.

**Assistant Sugar Maker** November 2013 – Present
*Howrigan Maple Heaven* *Fairfield, VT*
- Tap over 1,000 trees annually and maintain physical infrastructure including pressure filtration and tubing system leak repair.

**Retail Associate and Cafe Barista** October 2020 – February 2023
*Lake Champlain Chocolates* *South Burlington, VT*
- Delivered customer service while maintaining and stocking a high-volume retail sales floor.

**Legislative Page** March 2018 – May 2018
*State of Vermont* *Montpelier, VT*
- Supported Vermont State Senators and Representatives on the floor during the legislative session, tracking bills from introduction through passage.

## SKILLS

**Software and Tools:** NASTRAN, MATLAB, Simulink, DAQFactory, Python, NumPy, pandas, Autodesk Inventor, AutoCAD, Arduino/C++, Excel